

Tim Keller, PhD

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For more details, particularly regarding projects, please refer to
🔗 <https://timkeller15.github.io>.

PROFILE

Theoretical physicist with strong analytical and numerical skills specialized in the modeling and simulation of ultracold quantum gases as well as quantum optics and expertise in high-performance computing, including general-purpose GPU computing.

Independently planned and executed the major research directions in my projects that led to the steady output of findings published in peer-reviewed journals, including the modeling, numerical simulation, analytical calculations, data analysis, and writing of reports, while closely coordinating with collaborators for the interpretation of results.

EXPERIENCE

- **Junior Researcher** since Apr 2023
Waseda University, Tokyo, Japan
 - funded by JST Moonshot R&D Goal 6 Project: *Large-scale quantum hardware based on nanofiber cavity QED* led by 🧑‍🔬 Prof. Takao Aoki
 - developed theoretical models for applying quantum computing performance benchmarks to the lab's experimental platform of neutral atoms trapped next to optical nanofibers, providing design guidelines for optimizing quantum gate fidelities in this system
 - wrote a custom GPU-accelerated FDTD software package for numerically solving Maxwell's equations and calculating the atomic trapping potential generated by highly-focused laser beams reflected off a nanofiber.
- **JSPS Postdoctoral Fellow** Sep 2022 - Mar 2023
OIST, Okinawa, Japan
 - continuation of PhD work on the control of interacting quantum gases
 - supervision and training of 3 PhD students and 2 research interns

EDUCATION

- **PhD in Physics** Sep 2017 - Aug 2022
Thesis: 📄 Controlling Superfluid and Insulating States in Interacting Quantum Gases
Okinawa Institute of Science and Technology (OIST), Japan **Advisor:** Prof. Thomas Busch
- **MSc in Physics** Oct 2014 - Dec 2016
Saarland University, Germany **Advisor:** Prof. Giovanna Morigi
- **BSc in Physics** Oct 2011 - Aug 2014
Saarland University, Germany **Advisor:** Prof. Giovanna Morigi

FELLOWSHIPS & AWARDS

- **Research Fellowship for Young Scientists (DC2)** Apr 2021 - Mar 2023
Japan Society for the Promotion of Science **Grant:** 🧑‍🔬 JSPS KAKENHI 21J10521
Stipend: JPY 200 000 (~ USD 1800) per month & Research Grant of JPY 900 000 (~ USD 8200)

- **Outstanding Student Award** **Apr 2021 - Mar 2023**
Okinawa Institute of Science and Technology Graduate University
 Annual tuition of JPY 540 000 (~ USD 4800) waived in recognition of JSPS fellowship
- **Physical Review Letters Editor's Suggestion** **Jan 2022**
American Physical Society **Publication:**  PRL **128**, 053401 (2022).
 Flagship journal with acceptance rate < 25%. Only one in seven Letters highlighted as Editor's Suggestion due to their particular importance, innovation, and broad appeal.

PROJECTS & PUBLICATIONS

- (9) **Hikari.jl - GPU-accelerated FDTD Maxwell equation solver for calculating laser-generated atomic trapping potentials close to optical nanofibers**
 Contribution: *T. Keller* Code:  Hikari.jl
- (8) **Addressing requirements for crosstalk-free quantum-gate operation in many-body nanofiber cavity QED systems**
 Contribution: *T. Keller*, *S. Kikura*, *R. Asaoka*, *Y. Suzuki*, *Y. Tokunaga*, and *T. Aoki*
 Publication:  arXiv:2509.08408 (2025). Code:  AddressingRequirements.jl
- (7) **Fermionization of a few-body Bose system immersed into a Bose-Einstein condensate**
 Contribution: *T. Keller*, *T. Fogarty*, and *T. Busch*
 Publication:  SciPost Phys. **15**, 095 (2023). Code:  Fermionization
- (6) **Self-Pinning Transition of a Tonks-Girardeau Gas in a Bose-Einstein Condensate**
 Contribution: *T. Keller*, *T. Fogarty*, and *T. Busch*
 Publication:  Phys. Rev. Lett. **128**, 053401 (2022). (Editor's Suggestion) Code:  Self-Pinning
- (5) **Adiabatic critical quantum metrology cannot reach the Heisenberg limit even when shortcuts to adiabaticity are applied**
 Contribution: *K. Gietka*, *F. Metz*, *T. Keller*, and *J. Li*
 Publication:  Quantum **5**, 489 (2021).
- (4) **Feshbach engine in the Thomas-Fermi regime**
 Contribution: *T. Keller*, *T. Fogarty*, *J. Li*, and *T. Busch*
 Publication:  Phys. Rev. Research **2**, 033335 (2020). Code:  Feshbach-Engine
- (3) **Quenches across the self-organization transition in multimode cavities**
 Contribution: *T. Keller*, *V. Torggler*, *S. Jäger*, *S. Schütz*, *H. Ritsch*, and *G. Morigi*
 Publication:  New J. Phys. **20** 025004 (2018). Code:  MultimodeSelforganization.jl
- (2) **Phases of cold atoms interacting via photon-mediated long-range forces**
 Contribution: *T. Keller*, *S. Jäger*, and *G. Morigi*
 Publication:  J. Stat. Mech. 064002 (2017). Code:  MultimodeSelforganization.jl
- (1) **Probing the out-of-equilibrium dynamics of two interacting atoms**
 Contribution: *T. Keller* and *T. Fogarty*
 Publication:  Phys. Rev. A **94**, 063620 (2016). Code:  TwoAtomQuenches.jl

LANGUAGES

German: Native | **English:** Fluent (TOEFL Score 114/120) | **Japanese:** Intermediate (JLPT N2 Certificate)